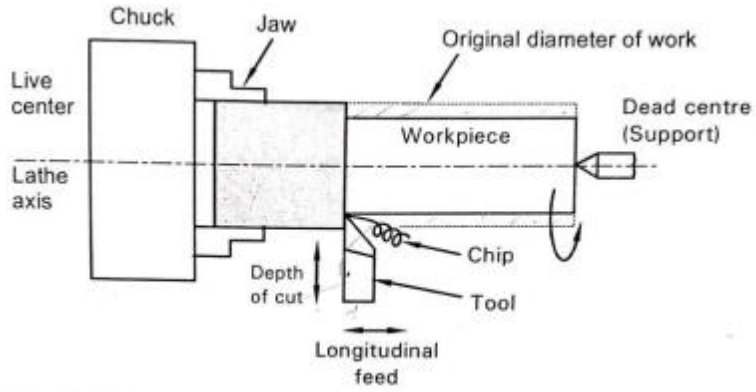
Machine Tool Operations

Lathe Machine:

A Lathe is defined as a machine tool used to remove excess material by forcing a cutting tool against a rotating work piece. Lathes are also called as Turning Machines, as the work piece is turned or rotated between two centers. They are primarily used to produce cylindrical, plain and tapered surfaces and used for knurling and thread cutting on metal parts. Besides these operations, they can also be used to perform various operations such as drilling, boring, reaming, grinding, etc.

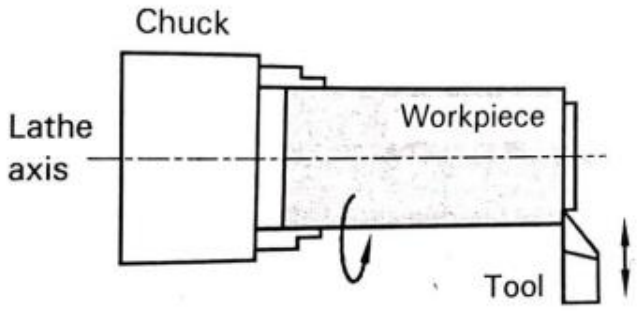
Lathe Operations:

Cylindrical Turning:



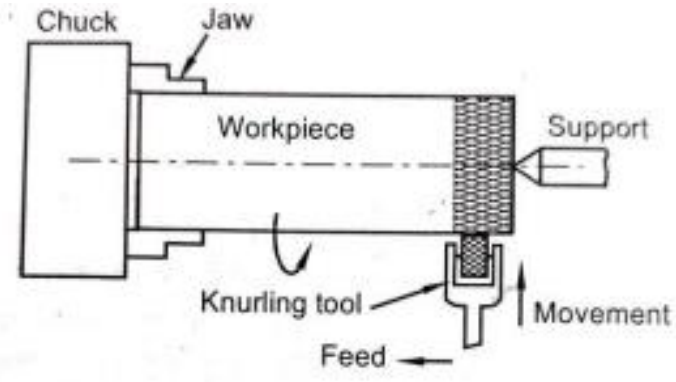
It is also called as plain turning or straight turning which is the operation of removing excess material from the work piece to produce a cylindrical surface of required diameter. The work piece is supported in between the two centers which permit the rotation of the work piece. A single point cutting tool is fed perpendicular to the axis of work piece to a known predetermined depth of cut and is moved parallel to the axis of the work piece.

Facing:



Facing is an operation performed on lathe machine to generate either flat surfaces or shoulders at the end of the work piece. Here the tool is moved perpendicular to the axis of the work piece to obtain the required dimension. The length of the work piece can be reduced in steps by performing lathe operation.

Knurling:



It is the process of embossing diamond shaped pattern on the surface of the work piece. It provides an effective means of gripping the work piece while handling. In knurling, the knurling tool is fed perpendicular to the axis of a slowly turning work piece which produce diamond pattern on it.

Working Principle of Drilling Machine

• When the power is given to the motor, the spindle rotates

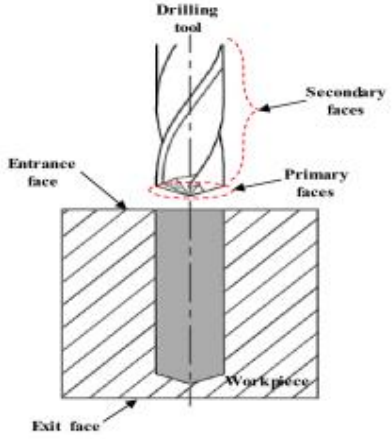
• Now the drill bit also rotates which was placed in the chuck and which was in connection with the spindle.

• As the Pulleys rotates, the spindle also rotates which can rotate the drill bit.

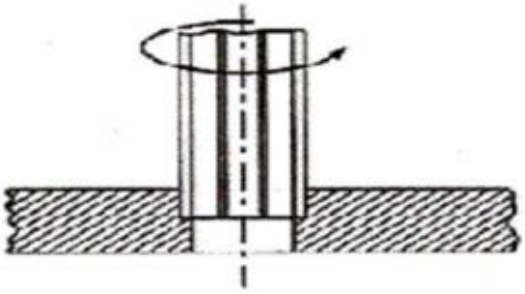
• Now, by the rotation of hand-wheel, the spindle moves up and down in the vertical direction to give the necessary amount of feed to the work and this drill bit is used to make the holes on the component placed in the machine vice.

Drilling:

Drilling is a cutting process where a drill bit is spun to cut a hole of circular cross-section in work piece. The drill bit is usually a rotary cutting tool, often multi-point. The bit is pressed against the workpiece and rotated at rates from hundreds to thousands of revolutions per minute.

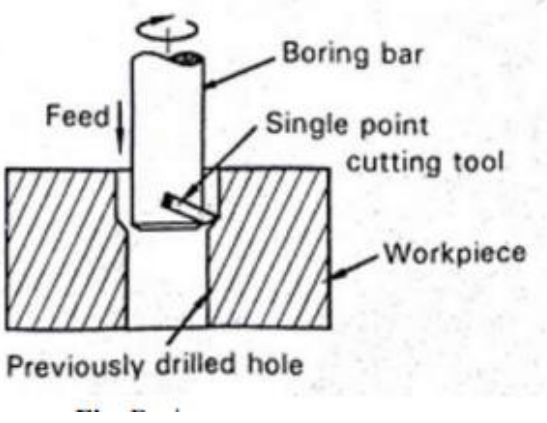


Reaming :



The holes that are produced by drilling are rarely straight and cylindrical in form. To produce accurate and smooth holes, the drilled holes are reamed by a tool called reamer. The reamer is a cutting tool having several cutting edges in straight or helix shape.

Boring:



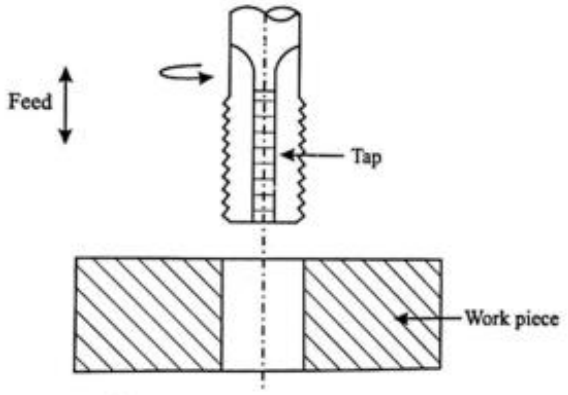
• Boring is a machining process carried out for enlarging a previously drilled hole by means of an adjustable cutting tool having only one cutting edge. shown figure below.

• Boring is usually performed when a drill bit of the required dimension is not available.

• In such cases, a hole is first drilled to the nearest dimension and then a single point cutting tool is fastened and adjusted to a boring bar to enlarge the size of the existing hole to the required dimension.

• While boring, the tool is rotated at speeds slower than that of reaming.

<u>Tapping</u>:



- The tapping is the process of cutting internal threads with a thread cutting tool called tap.
- A tap is a fluted threaded tool used for cutting internal threads. Before tapping, a hole which is slightly smaller than the size of the tap is drilled. For cutting the threads, the tap is fitted in the tapping attachment which in turn is mounted in the drilling machine spindle, and the threads are cut in the same way as drilling.
- While tapping in a drilling machine the spindle must rotate at very slow speeds.

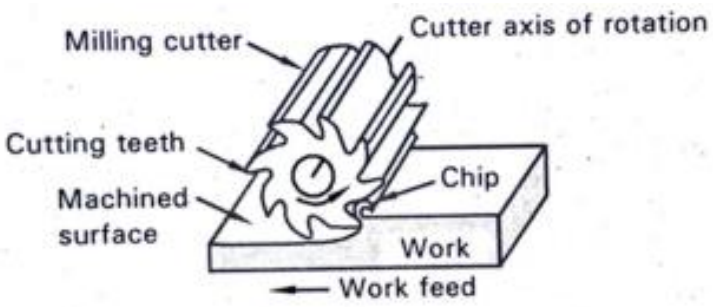
INTRODUCTION TO MILLING

A milling machine is a power operated machine tool in which the work piece mounted on moving table is machined to various shapes when moved under a slow revolving multipoint cutter.

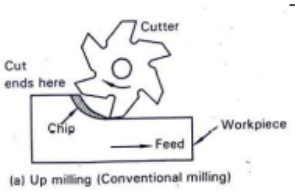
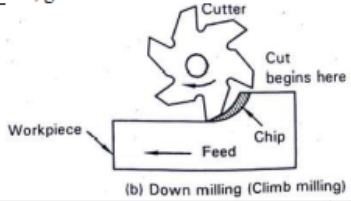
• The milling cutter is a multipoint cutting tool. The work piece is mounted on a movable worktable which will be fed against the revolving milling cutter to perform the cutting operation

• The difference between drilling and milling is that, in drilling a rotating drill is fed against a stationary work piece, while in milling the work piece is fed against a milling cutter which only revolves.

• Similarly, it also differs from the lathe operation because the lathe tool is fed against rotating work piece.

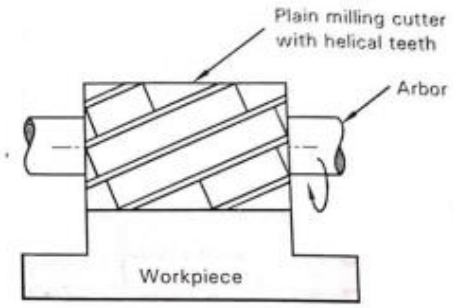


COMPARISON BETWEEN UP MILLING AND DOWN MILLING

Sl. No.	Up milling (Conventional milling)	Down milling (Climb milling)
1.	In up milling, the workpiece is fed in the direction opposite to that of the rotating cutter.	In down milling, the workpiece is fed in the same direction as that of the rotating cutter.
2.	The thickness of chip is minimum at the beginning of cut and reaches to a maximum when the cut ends.	The thickness of chip is maximum at the beginning of cut and reaches to the minimum when the cut ends.
3.	In up milling, the cutting force is directed upwards. This tends to lift the workpiece from the worktable. Hence, greater clamping force for the workpiece becomes necessary.	The cutting force is directed downwards, and this tends to keep the workpiece firmly on the worktable thereby permitting lesser clamping forces.
4.	During up milling, the chip gets accumulated at the cutting zone (tool-work interface). These chips interfere with the rotating cutter thereby impairing the surface finish on the work-surface.	In down milling, the chips do not interfere with the revolving cutter, since they are disposed easily by the cutter. Hence, there is no damage to the surface finish of the workpiece.
5.	In up milling, it is difficult for efficient circulation of coolant. The cutter rotating in the upward direction carries away the coolant from the cutting zone.	In down milling, the coolant can easily reach the cutting zone. Hence, efficient cooling of the tool and the workpiece can be achieved.
6.	Up milling is preferred for rough cuts, especially for castings and forgings, because this method enables the cutter to dig-in and start the cut below the hard upper surface.	Down milling produces better surface finish because there is no dig-in of the cutter. It is particularly used for finishing operations and small work like cutting slots, grooves etc.
7.	 (a) Up milling (Conventional milling)	 (b) Down milling (Climb milling)

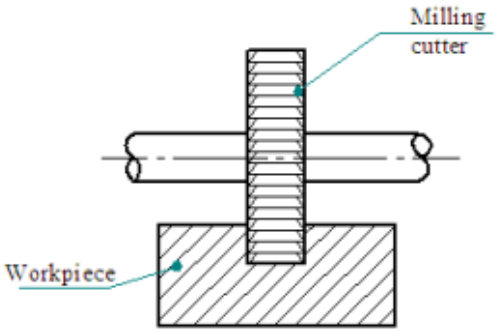
Milling Operations

Plain Milling



Also called as slab milling is a process used to mill flat surfaces of work piece in such a way that the milling cutter axis is parallel to the surface that is being milled.

Slot Milling:

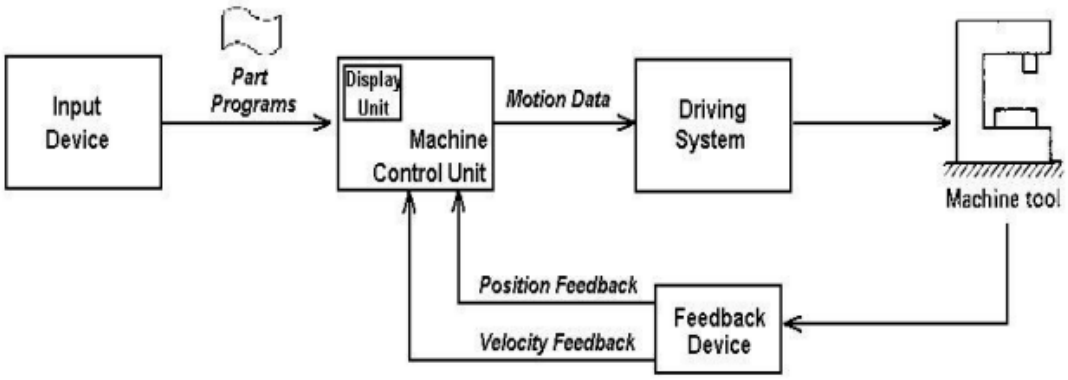


It is a process to produce desired slots, grooves and keyways using suitable cutters. Figure illustrates groove milling operation using a milling cutter, which has teeth on its periphery. Similarly open slots and closed slots can be cut by using plain milling cutters and end milling cutters respectively.

Basic Components of CNC Machine

CNC Machine basically consists of three components:

1. Part Program and Program Input Device
2. Controller unit or Machine Control Unit (MCU)
3. Machine Tool



Part Program and Program Input Device: In a CNC machine, the part program is done either manually on punched tapes and then fed into the memory of the computer or with the aid of various computer software that automatically generates the program required to produce the desired part.

Machine Control Unit (MCU): In modern CNC technology, the machine control unit consists of a microcomputer and related control hardware that stores the program of instructions and executes it by converting each command into mechanical actions of machine tool, one command at a time. The MCU includes system software, calculation algorithm and transition software to convert the part program into a usable format for the MCU. It also acts as a feedback controller for precise positioning of machine table or spindle.

Machine Tool: The machine tool performs the actual machining operation and can be of any type like lathe, drilling machine, milling machine, etc. The machine tool mainly consists of worktable and tool spindles whose motion can be controlled accurately by the driving system that receives commands from the MCU.

Advantages of CNC

- CNC machines can be used continuously 24 hours a day, 365 days a year.

• CNC machines are programmed with a design which can then be manufactured hundreds or even thousands of times.

• Each manufactured product will be the same.

• Modern design software allows the designer to simulate the manufacturing, as there is no need to make a prototype or a model. This saves time and money.

• A person can control many CNC machines as once programmed, they can do the tasks efficiently without human intervention.

Disadvantages of CNC

• Expensive
• Requires programming skills
• Requires skills for setting up the machines
• Leads to unemployment
• Software compatibility issues can cause some programs to run in one machine, which can/cannot run in another

Applications of CNC

• Embroidery machines
• Woodcutters
• Wire-bending machines
• Foam cutters
• Laser cutters

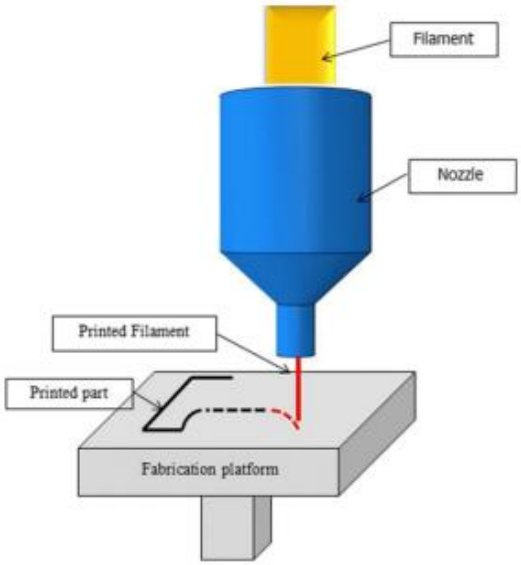
3D printing Process

3D printing, commonly known as Additive manufacturing creates 3-dimensional objects by successively layering materials under computer control. 3D printing involves layering materials like plastics, composites, or biomaterials to create objects that range in shape, size, rigidity, and color.

Working process of a 3D printer:

- **3D Model Design -** The process begins with creating a 3D model of the object using computer-aided design (CAD) software. The model is sliced into layers.
- **Machine Preparation -** The 3D printer is loaded with the required raw materials like plastic filament, resin, metal powder etc. based on the printing technology used.
- **Printing Materials -** Unlike inkjet and laser printers, a 3D printer does not use liquid ink or solid powder. It deposits layers of molten plastic, metal wire, or powder and fuses them with the existing structure using adhesives or ultraviolet light.

- **Layer-by-Layer Printing -** The 3D printer deposits and fuses together material layer-by-layer as per the design file. The build platform lowers, and the process is repeated until the object is formed.
- **Post-Processing -** Printed objects may require additional steps like cleaning, smoothening, baking, and polishing. Support structures are removed.
- **Quality Checks -** Printed objects are checked for dimensional accuracy and compliance with design specifications. Defects may require re-printing.
- **Application -** The final 3D printed part can then be put to functional use or for other applications like prototyping, education, art and so on.
- **Materials:** A range of materials are used for 3D printing including plastics like ABS, PLA, nylon, photopolymers, metals including titanium, aluminum, steel, gold, silver, alloys, ceramics, paper, wax, and even food materials like chocolate.



Basic 3D Printer Layout:

• **Print Bed**: The print bed is the surface that your objects are printed on to. Typically, it will consist of a sheet of glass, a heating element, and some kind of surface on top to help the plastic stick.

Filament: This is the plastic that's consumed by the printer. It comes on a spool.

• Extruder: The extruder is the core of the printer. It is where the plastic gets drawn in, melted, and pushed out.

Heater Cartridge: It heats the plastic.

Thermistor/Thermocouple/RTD: These are all various types of sensors for determining the temperature of the hot end

Nozzle: The nozzle is simply a piece with a small hole for the melted filament to come out of. Nozzles are interchangeable and come in various sizes

Stepper Motors: Stepper motors rotate in increments and gives precise control over their position.

Power Supply: This takes the 120V AC electricity from the wall and converts it to low voltage DC power.

Controller Board: It takes the commands given to it by your computer and orchestrates their execution. It contains a microcontroller and all the circuitry needed for running the motors, reading the sensors, and talking to your computer.

Applications of 3D Printing

- **Aerospace:** Lightweight printed aircraft and spacecraft parts can replace traditionally manufactured parts to reduce weight and material needs.
- **Automotive**: The automotive industry uses 3D printing for rapid prototyping and to produce specialized parts and tools. Printed parts help improve design and can reduce weight.
- **Construction:** 3D printing technology is being used to construct scale models of buildings, bridges, and even full-scale houses and other structures using concrete and other materials.
- **Healthcare**: 3D printing is revolutionizing healthcare with customized prosthetics, implants, anatomical models for surgery practice, and pharmaceuticals.
- **Consumer**: With 3D printing, products can be easily customized or personalized to an individual's specs.
- **Public Infrastructure**: India's first 3D printed post office built in just 45 days by L&T and IIT Madras demonstrates rapid construction capabilities.

Challenges of 3D Printing: Speed, Cost, Accuracy, Surface finish, Strength, Material options

Questions:

1. With neat sketch explain the following machine tool operations i) Boring ii) Plane milling iii) Slot milling
2. Explain the components of CNC with block diagram.
3. What are the advantages and applications of CNC?
4. Write a short note on 3D printing
5. Explain the following lathe operations with neat sketch: i) Turning ii) Knurling
6. Illustrate the following operations of drilling with sketches. i) Boring ii) Drilling iii) Reaming
7. Explain the working principle of Milling with schematic diagram
8. Define Additive Manufacturing. List the various steps involved in additive manufacturing
9. List the advantages and applications of CNC